

Indian Food Nutrient Profiling Engine: Algorithm, Methodology & Standards

1. Executive Summary & Clinical Rationale

NutriScore AI is an evidence-based front-of-pack (FoP) nutritional profiling system calibrated specifically for traditional and contemporary Indian recipes. It establishes an objective, transparent, and clinically rigorous scoring mechanism that benchmarks foods against the **Indian Council of Medical Research (ICMR-NIN 2020) Recommended Dietary Allowances** and the **Anuvaad Indian Nutrient Databank (INDB)**.

Why Western Nutri-Score Fails in India:

- **Omits Critical Micronutrients:** Traditional European Nutri-Score (FSA-NPS) measures only 4 negatives and 3 positives, giving zero credit to Iron, Calcium, Vitamin A, Vitamin C, or Potassium—the primary nutritional deficiencies affecting the Indian subcontinent.
- **Unfair Penalty on Pulses (Dals):** European rules cap protein credit unless a recipe contains >80% fruit/vegetables, penalizing healthy lentil dishes.
- **NutriScore AI Solution:** Implements a dual-step balanced algorithm (5 negative penalties, max 50 pts; 11 positive rewards, max 70 pts) normalized onto a continuous 0 to 100 Health Score and 5-color A to E grade.

2. Step 1: Negative Nutrient Scoring (N) — Nutrients to Limit

Negative nutrients correlate with chronic non-communicable diseases (NCDs) such as type-2 diabetes, hypertension, coronary heart disease, and obesity. In the primary **ICMR 16-Nutrient Model**, there are **5 negative nutrients**, each penalized from 0 to 10 points based on progressive cutoff thresholds (Max N = 50).

Negative Nutrient	Unit	Max Pts	Optimal (0 pts)	Severe (10 pts)	Clinical Rationale & Disease Impact
Energy Density	kcal/100g	10	≤ 80	> 800	Excess caloric density directly drives adiposity and metabolic syndrome.
Free Sugars	g/100g	10	≤ 4.5	> 45.0	Triggers acute glycemic spikes, hepatic de novo lipogenesis & insulin resistance.
Saturated Fat	g/100g	10	≤ 1.0	> 10.0	Elevates serum LDL-cholesterol and atherosclerotic cardiovascular risk.
Sodium (Salt)	mg/100g	10	≤ 90	> 900	Primary dietary determinant of systemic hypertension, stroke, and kidney strain.
Cholesterol	mg/100g	10	≤ 20	> 200	Accelerates vascular plaque build-up and coronary artery calcification.

3. Step 2: Positive Nutrient Scoring (P) — Nutrients to Encourage

Positive scoring recognizes essential macros, slow-digesting complex carbohydrates, healthy lipids, and vital micronutrients calibrated directly against the **ICMR-NIN 2020 RDA for Indians**. Every 5% of daily adult RDA awards 1 point, reaching the maximum 5 points at ≥ 25% RDA (Max P = 70).

Positive Nutrient	Category	ICMR RDA	Max Pts	Max Cutoff (5 pts)	Nutritional Role in Indian Diet
Protein	Macro	54 g	5	> 8.0 g	Cellular repair, enzymatic function, muscle mass preservation.
Dietary Fibre	Macro	30 g	5	> 4.7 g	Microbiome health, satiety, and blunt postprandial glucose spikes.

Complex Carbs	Macro	Derived	5	> 25.0 g	Sustained glycemic energy (derived: Carbs - Free Sugars - Fibre).
MUFA & PUFA	Lipid	30 g	5 + 5	> 5.0 g each	Essential unsaturated fatty acids; cardioprotective lipid ratios.
EAAI Quality Bonus	Protein	ICMR Index	10	Index ≥ 0.90	Rewards complementary amino acid profiles (e.g., Dal + Roti).
Iron (Fe)	Micro	19 mg	5	≥ 4.75 mg (25% RDA)	Hemoglobin synthesis; addresses widespread maternal & child anemia.
Calcium (Ca)	Micro	1,000 mg	5	≥ 250 mg (25% RDA)	Skeletal mineral density, neuromuscular signaling, cardiac pacing.
Vitamin A	Micro	1,000 mcg	5	≥ 250 mcg (25% RDA)	Vision cycle integrity, epithelial immunity, gene transcription.
Vitamin C	Micro	80 mg	5	≥ 20 mg (25% RDA)	Potent antioxidant; triples bioavailability of non-heme plant iron.
Potassium (K)	Micro	3,500 mg	5	≥ 875 mg (25% RDA)	Intracellular osmolarity; counteracts vascular strain caused by sodium.

4. Mathematical Formulation & Normalization (0–100)

Unlike legacy systems that output arbitrary negative integers (e.g. -5 to +35), NutriScore AI normalizes the net balance between positive rewards (P) and negative penalties (N) onto an intuitive 0 to 100 percentage scale:

$$\text{Health Score} = 100 \times [(P - N + N_{\text{max}}) / (P_{\text{max}} + N_{\text{max}})]$$

For the ICMR 16-Nutrient Model: $N_{\text{max}} = 50$, $P_{\text{max}} = 70$, Total Scale Span = 120

$$\text{Health Score} = 100 \times [(P - N + 50) / 120]$$

5. 5-Color Nutri-Score Grade Classification (A to E)

The continuous Health Score (0–100) is classified into 5 standardized tiers recognized internationally:

Grade	Score Range	Color Code	Nutritional Interpretation & Dietary Guidance
Grade A	75 – 100	■ Dark Green	Outstanding Nutritional Quality: Dense in protein, fibre, and essential micronutrients. Negligible free sugars and saturated fats. Highly encouraged for daily staple consumption.
Grade B	60 – 74	■ Light Green	Good Nutritional Quality: Favorable nutrient density with balanced energy. Optimal as core components of everyday balanced Indian meals (e.g. Tadka Dal, Khichdi).
Grade C	45 – 59	■ Yellow	Moderate Nutritional Quality: Standard baseline food. Provides calories and basic macronutrients but contains modest sodium or fats and lacks top-tier micronutrient density.
Grade D	30 – 44	■ Orange	Sub-optimal / Caution: High in one or more risk nutrients (excess sodium, frying oil, or refined carbs). Recommend reducing portion sizes and frequency of intake.
Grade E	0 – 29	■ Red	Poor Nutritional Quality: Ultra-energy-dense, high saturated fats, refined sugar syrups, or excess sodium with virtually zero protective fibre or micronutrients (e.g. Gulab Jamun, deep-fried snacks).

6. Case Studies: Algorithm Evaluation on Indian Recipes

Recipe	Key Nutrient Profile (per 100g)	Negative (N)	Positive (P)	Health Score	Resulting Grade
Moong Dal Tadka (Lentil Curry)	120 kcal, 7.5g Protein, 4.2g Fibre, 0.5g Sugar, 0.8g Sat Fat, 180mg Na, 2.1mg Fe, 45mg Ca	4 / 50 pts	28 / 70 pts	61.7 / 100	Grade B (Good)
Gulab Jamun (Syrup Sweet)	380 kcal, 38.0g Free Sugar, 9.5g Sat Fat, 2.8g Protein, 0.2g Fibre, 45mg Cholesterol	36 / 50 pts	3 / 70 pts	14.2 / 100	Grade E (Poor)

7. Profiling Frameworks Comparison

NutriScore AI provides three selectable profiling algorithms to support both everyday consumer guidance and advanced biochemical research:

Framework Model	Nutrient Scope	Penalty (N)	Reward (P)	Scale Span	Target Application & Clinical Context
ICMR 16-Nutrient (Default & Core)	16 Core INDB Components	Max 50 pts (5 nutrients)	Max 70 pts (11 nutrients)	Span = 120 (0 to 100)	Consumer Standard: Calibrated for traditional Indian meals; realistic distribution across B, C, and D tiers.
Extended 39-Nutrient (INDB Research)	All 39 Lab Components	Max 50 pts (5 nutrients)	Max 165 pts (34 nutrients)	Span = 215 (0 to 100)	Laboratory & Clinical: Incorporates 8 trace minerals and 10 B-complex vitamins for full biochemical profiling.
Official Nutri-Score (FSA-NPS Europe)	7 Packaged Components	Max 40 pts (4 nutrients)	Max 15 pts (3 nutrients)	Score: N - P (-15 to +40)	International Benchmark: Direct implementation of French Santé Publique standard for cross-continental comparison.

8. Technical Architecture & Live Deployment

- **Asynchronous REST Backend:** Python 3.11+ running FastAPI and Uvicorn. Implements strict Pydantic v2 validation contracts, dynamic CORS middleware, and automated unit testing (18/18 passing test suite).
- **Reactive Single-Page Client:** React 18, TypeScript, Vite, and Tailwind CSS. Built with responsive mobile-first ergonomics, Recharts data visualization radar graphs, and real-time client-side input validation.
- **Production Cloud Hosting:**
 - **Backend API:** Deployed on **Render.com** (<https://nutriscore-india.onrender.com>) with automated healthcheck keep-alive via cron-job.org.
 - **Frontend UI:** Deployed globally on the **Vercel Edge Network** with custom SPA rewrite rules.
 - **Open-Source Repository:** Publicly hosted on GitHub at <https://github.com/HG12265/NutriScore-India>.